Fuselage Section 13 — Forward — Door Surround Structure

Fwd Cargo Door Surround Fitting — Corner Crack Stop Drill and Reinforcing Angle

Object code	DMC-SLN-A-53-21-13-00A-664A-A	**Issue**	002-00 (revised)
Issue date	2026-07-30	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Repair Design, Neuhafen (U8025)
Security	01 — Unclassified	**Quality assurance**	First verification (tabtop)
Applicability	SOLANO-30 and SOLANO-34, all aircraft		

1. Reason for the job

This repair arrests a crack at the lower corner of the forward cargo door surround fitting and restores enough static capability for continued operation until the fitting is replaced at the next opportunity.

2. Job set-up information

A. Referenced information

TYPE	DESIGNATION	IDENTIFICATION No.
GEN	Cracks up to 18 mm measured from the radius tangent point, originating at the corner radius, not crossing a fastener row and not branched. One crack per fitting.	—
GEN	The damage is in zone 111, 121.	—
GEN	A repair approval request has been raised with Aeralis.	—

B. Fixtures, tools, test and support equipment

No support equipment required

C. Consumable materials

QTY	DESIGNATION	IDENTIFICATION No.
36 EA	WASHER, COUNTERSUNK, ALUMINIUM	AES963502K7 (CAGE U8025)
36 EA	WASHER, PLAIN, CORROSION RESISTANT	AGS46790-6A (CAGE U8025)

D. Expendable parts

No expendable parts required

3. Procedure

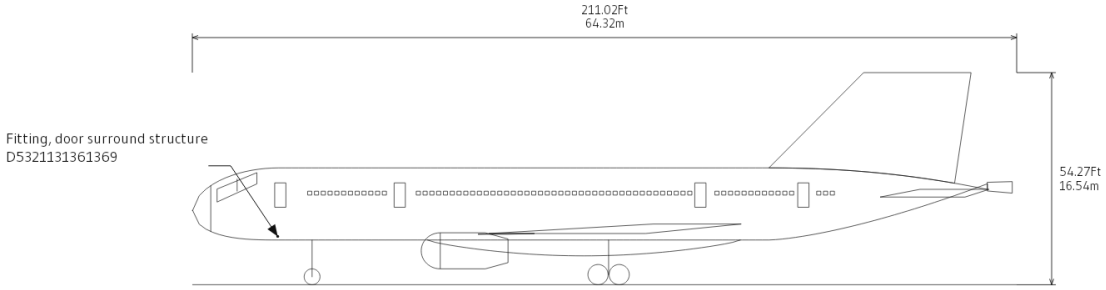
1. Damage assessment

Find the crack tip by dye penetrant inspection and mark it. Cracks up to 18 mm in length, originating at the corner radius and not crossing a fastener row, are inside this repair. A crack that has reached a fastener hole, that has branched, or that lies in the frame sub-assembly rather than in the fitting, is outside it and requires an engineering assessment.

3. Procedure

2. Measure the crack length from the radius tangent point, not from the visible surface break, because the subsurface extent is typically 3 to 5 mm longer. Do an eddy current inspection of the opposite corner of the same fitting before deciding on the repair; symmetrical cracking usually means the load path has changed and a single-corner repair will not hold. Record the protection standard of the surround at the same time; it is read from the aircraft configuration and confirmed by the presence of the interrupted sealant land, and it changes the layout of the repair.

3. See Fuselage Section 13 — Forward — Door Surround Structure — location, side view.



Fuselage Section 13 — Forward — Door Surround Structure — location, side view — Figure 1

4. Repair procedure

Stop drill the crack 4 mm beyond the marked tip with a 4.8 mm drill, and ream the hole to remove the drilled surface. Do a dye penetrant inspection of the stop drill hole to confirm the crack tip is contained. Cut the reinforcing angle from the fitting kit D5321131361369 material to the layout of repair drawing RD-53-21-13-003 and drill the fastener pattern using the angle as a template.

5. Establish the protection standard of the surround before the template is positioned. On the original standard the sealant land runs the full length of the frame foot and the inboard row is set out 12 mm from it. On the chromate-free standard the land is interrupted and its inboard edge sits 9 mm further in, so the row is set out 16 mm from the remaining land: a 12 mm offset on that standard puts the row into the drain interruption and the two dimensions cannot both be met.

6. Do not drill into the frame sub-assembly D5321134517524; the drawing gives the exclusion boundary. Apply the chromate-free primer to all new surfaces. Install the angle wet with sealant and fasten it with washers AES963502K7 and plain washers AGS46790-6A at the pitch on the drawing.

7. Fit silicone edging grommets where the angle crosses a systems penetration. Restore the sealing at the frame foot, keeping the interrupted land clear on aircraft that have it. The angle must not bridge the interruption; where the layout would close it, mask the land and leave the angle short.

8. Inspection requirements

This repair is interim and is limited to 6 000 flight cycles or twenty-four months from embodiment, whichever occurs first, after which the fitting must be replaced. Do a repeat dye penetrant inspection of the stop drill hole and of the angle perimeter every 1 500 flight cycles. Any growth from the stop drill hole makes the repair invalid immediately.

9. Record the repair against the aircraft and notify Aeralis within ten days.

10. Repair data

Table 1 Repair data

Repair number	53-21-13-003
Category	Interim
Damage type	Crack
Fastener pattern	Two rows of 18 fasteners at 18 mm pitch, staggered. Inboard row edge distance is 12 mm from the sealant land on the original protection standard and 16 mm on the chromate-free standard; the exclusion boundary on sheet 2 is left unfastened
NDT method	Dye penetrant
Static strength restored	85 %
Mass penalty	0.9 kg
Aerodynamic smoothness	Not applicable, internal repair
Aeralis approval required	Yes
Repair drawings	RD-53-21-13-003 (2 sheets)

4. Close-up

Interim repair limited to 6 000 flight cycles or twenty-four months; repeat dye penetrant inspection every 1 500 flight cycles.

Restore the external protection scheme and apply the repair stencil.